Honors Calculus III, First midterm

This exam consists of two problems for a total of 30 points. Attempt to solve as many problems as you can. Make sure you read each question carefully and understand what you are being asked to do. You should progress sequentially, justify each step, use notation correctly, and clearly mark the final answer. Please write your name and section number on every answer booklet you use.

In this exam you can use results from lectures or homework, or Math 16100/16200, provided you state them clearly and you avoid circular arguments: for example, you cannot use a result that you are explicitly asked to prove in the question. As usual, for full credit you need to show and explain your work.

Problem 1.

1. (7pt) Let f_n be a sequence of continuous functions on $[0,1]$. Assume that $f_n \to f$ uniformly on $[0,1]$. Prove that f is continuous on $[0,1]$.
2. (4pt each) Determine (with proof) whether the following sequences of functions converge uniformly or not. You can assume without proof that they converge pointwise to zero.

$$f_n(x) = \frac{1+\sin x}{n+x^2} \text{ for } x \in [0,1]$$
$$g_n(x) = \sqrt{x^3 + \frac{1}{n^3}} - \sqrt{x^3} \text{ for } x \in [1,+\infty).$$

Problem 2. Consider the sequence $\{\alpha_n\}$ defined recursively by

$$\alpha_0 = 2, \alpha_1 = 1 \text{ and } \alpha_{n+2} = \alpha_{n+1} + \alpha_n$$

for every $n \geq 0$.

1. (5pt) Prove that the power series
$$F(x) = \sum_{n\geq 0} \alpha_n x^n$$
converges absolutely for $|x| < 1/2$.
2. (5pt) Assume that $|x| < 1/2$. Prove that
$$\sum_{n\geq 0} \alpha_n x^n = \frac{2-x}{1-x-x^2}.$$
3. (5pt) Let
$$\varphi = \frac{1+\sqrt{5}}{2} \text{ and } \overline{\varphi} = \frac{1-\sqrt{5}}{2}.$$
Using the partial fraction decomposition
$$\frac{2-x}{1-x-x^2} = \frac{1}{1-\varphi x} + \frac{1}{1-\overline{\varphi} x},$$
deduce the formula
$$\alpha_n = \varphi^n + \overline{\varphi}^n.$$
(You don't have to prove the decomposition.)

Solutions to Problem 1. (1) This is a theorem from class, see Spivak, Chapter 24, Theorem 2.

(2a) The sequence converges uniformly because for all $x \in [0,1]$ we have

$$|f_n(x)| \leq \frac{2}{|n+x^2|} \leq \frac{2}{n}$$

and so

$$\lim_{n\to+\infty} \sup_{x\in[0,1]} |f_n(x)| = 0.$$

This implies uniform convergence to zero by a theorem from class.

(2b) This also converges uniformly. Notice that

$$g_n(x) = \frac{1/n^3}{\sqrt{x^3+(1/n^3)}+\sqrt{x^3}}.$$

Since

$$\sqrt{x^3+(1/n^3)}+\sqrt{x^3} > \sqrt{1/n^3}$$

whenever $x > 0$, we have

$$|g_n(x)| \leq \frac{1/n^3}{\sqrt{1/n^3}} = \frac{1}{n^{3/2}} \to 0 \text{ as } n \to \infty.$$

It follows that

$$\lim_{n\to\infty} \sup_{x\in[1,+\infty)} |g_n(x)| = 0.$$

Solutions to Problem 2. By the ratio test (or rather its proof; see below) it suffices to prove that

$$|\alpha_{n+1}/\alpha_n| \leq 2.$$

for all n.

Using the recurrence relation, we have

$$\alpha_{n+2}/\alpha_{n+1} = \frac{\alpha_{n+1}+\alpha_n}{\alpha_{n+1}} = 1 + \frac{\alpha_n}{\alpha_{n+1}}.$$

By definition, $\alpha_{n+1} \geq \alpha_n$ for all $n > 0$, since then α_{n+1} is obtained from α_n by adding the positive number α_{n-1}. Hence we have

$$|\alpha_{n+2}/\alpha_{n+1}| \leq 2$$

for all $n > 0$, which implies the result.

Remark: The ratio test states that if

$$\lim_{n\to\infty} \frac{|\alpha_{n+1}x^{n+1}|}{|\alpha_n x^n|} < 1$$

then the series $\sum_{n\geq 0} \alpha_n x^n$ converges absolutely. In our case, it follows that if

$$\lim_{n\to\infty} |\alpha_{n+1}/\alpha_n| \leq 2$$

and $|x| < 1/2$ then $F(x)$ converges absolutely. If we prove that $|\alpha_{n+1}/\alpha_n| \leq 2$ for all n **and we prove that the limit** $\lim_{n\to\infty} |\alpha_{n+1}/\alpha_n|$ **exists**, then the limit will be at most 2, and so we will be done by the ratio test.

However, arguing as in the proof of the ratio test it is possible to bypass the need of proving existence of the limit. Assume that $|\alpha_{n+1}/\alpha_n| \leq 2$ for all n large enough, and that $|x| < 1/2$. Choose $\rho \in (|x|, 1/2)$. Then it follows that

$$\frac{|\alpha_{n+1}x^{n+1}|}{|\alpha_n x^n|} < 2\rho$$

for all large enough n. Since $|2\rho| < 1$, we deduce that $F(x)$ converges absolutely by comparison with the convergent geometric series $\sum_{n\geq 0}(2\rho)^n$.

In the context of this problem, you can also notice that $\alpha_{n-1} \leq \alpha_n$ for all $n \geq 2$, and so $\alpha_{n+1} = \alpha_n + \alpha_{n-1} \leq 2\alpha_n$. By induction, this implies that $\alpha_{2+i} \leq 2^i\alpha_2$. Now the series

$$\sum_{i\geq 0} 2^i x^i$$

converges absolutely by the ratio test if $|x| < 1/2$, and so $F(x)$ converges absolutely by the comparison test.

(2) Let $f(x) = \sum_{n\geq 0} \alpha_n x^n$, a function on the open interval $(-1/2, 1/2)$. It suffices to prove that

$$(1 - x - x^2)f(x) = 2 - x$$

The left-hand side is given by the power series

$$\alpha_0 + (\alpha_1 - \alpha_0)x + \sum_{n\geq 2}(\alpha_n - \alpha_{n-1} - \alpha_{n-2})x^2$$

and the recurrence relation implies that $\alpha_n - \alpha_{n-1} - \alpha_{n-2} = 0$ for all $n \geq 2$, so this simplifies to

$$\alpha_0 + (\alpha_1 - \alpha_0)x = 2 - x$$

since we are given that $\alpha_0 = 2$ and $\alpha_1 = 1$.

(3) We expand the fractions $(1 - \varphi x)^{-1}$ and $(1 - \overline{\varphi}x)^{-1}$ as geometric series:

$$\frac{1}{1 - \varphi x} = 1 + \sum_{n\geq 1} \varphi^n x^n$$
$$\frac{1}{1 - \overline{\varphi}x} = 1 + \sum_{n\geq 1} \overline{\varphi}^n x^n.$$

These expressions converge if $|x| \leq |1/\varphi|$ and $|x| \leq |1/\overline{\varphi}|$. If in addition $|x| < 1/2$, we find that the two power series

$$\sum_{n\geq 0} \alpha_n x^n \text{ and } \sum_{n\geq 0}(\varphi^n + \overline{\varphi}^n)x^n$$

define the same function of x. By a theorem from class, this implies that they have the same coefficients, which means that

$$\alpha_n = \varphi^n + \overline{\varphi}^n.$$